

$$\Delta(k^2) = \frac{1}{k^2 + m^2 - i\epsilon} + \int_{-\infty}^{\infty} ds \rho(s) \frac{1}{k^2 + s - i\epsilon}$$

$\rho(s)$: Spectral Density, $\in \mathbb{R}^+$

Also saw that

$$\Delta(k^2) = \frac{1}{k^2 + m^2 - i\epsilon - \pi(k^2)}$$

* Show how these relate for ϕ^3 at order $\alpha = g^2/(4\pi)^3$

* Take k^2 and m^2 to be real, then using $(x-i\epsilon)^{-1} = x/(x^2+\epsilon^2) + i\epsilon/(x^2+\epsilon^2)$

Proof:
$$\frac{x+i\epsilon}{(x-i\epsilon)(x+i\epsilon)} = \frac{x+i\epsilon}{x^2+\epsilon^2} = P\left(\frac{1}{x}\right) + i\pi\delta(x)$$

P: principal part of a f^n $f = f(z)$

is
$$P(f) = \sum_{i=1}^{\infty} a_{-i}(z-a)^{-i}$$

So knot ski - Plemelj Formula:

$$\lim_{\epsilon \rightarrow 0} \left(\frac{1}{x \pm i\epsilon} \right) = P\left(\frac{1}{x}\right) \mp i\pi\delta(x)$$

Proof: $\left(\frac{1}{x \pm i\epsilon} \right) = \frac{x \mp i\epsilon}{x^2 + \epsilon^2}$, $0 < \epsilon < 1$

Now, let $f(x) \in C_0^\infty(\mathbb{R})$ be a smooth f^n

$\int_{-\infty}^{\infty} \frac{f(x)}{x \pm i\epsilon} dx = \int_{-\infty}^{\infty} \frac{f(x)x}{x^2 + \epsilon^2} dx \mp i\epsilon \int_{-\infty}^{\infty} \frac{f(x)}{x^2 + \epsilon^2} dx$

$$\int_{-\infty}^{\infty} \frac{f(x)x}{x^2 + \epsilon^2} dx = \lim_{\delta \rightarrow 0} \left(\int_{-\infty}^{-\delta} \frac{f(x)x}{x^2 + \epsilon^2} dx + \int_{-\delta}^{\delta} \frac{f(x)x}{x^2 + \epsilon^2} dx + \int_{\delta}^{\infty} \frac{f(x)x}{x^2 + \epsilon^2} dx \right)$$



Note we can take the limit $\varepsilon \rightarrow 0$ for $\varepsilon =$

since $1/0$ is not a singularity

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{f(x)x}{x^2 + \varepsilon^2} dx = \lim_{\delta \rightarrow 0} \left(\int_{-\delta}^{\delta} \frac{f(x)x}{x^2 + \varepsilon^2} dx + \int_{-\infty}^{-\delta} \frac{f(x)x}{x^2} dx + \int_{\delta}^{\infty} \frac{f(x)x}{x^2} dx \right)$$

Since $\delta \ll 1$ then, since f is not singular near $x=0$

$$f(x) = f(0) + x f'(0) + \dots$$

$$f(\varepsilon) = f(0) + \varepsilon f'(0) + O(\varepsilon^2)$$

So if $|x| < \delta \ll 1$ then

$$f(x) \approx f(0) \text{ in } x \in [-\delta, \delta]$$

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{f(x)x}{x^2 + \varepsilon^2} dx = \lim_{\delta \rightarrow 0} \left(\int_{-\delta}^{\delta} \frac{f(0)x}{x^2 + \varepsilon^2} dx + \int_{-\infty}^{-\delta} \frac{f(x)}{x} dx + \int_{\delta}^{\infty} \frac{f(x)}{x} dx \right)$$

$\Rightarrow$ Since this is an odd f^n over a symmetric interval

* Now define the Principal value int.

$$P \left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx \right) = \lim_{\delta \rightarrow 0} \left\{ \int_{-\infty}^{-\delta} \frac{f(x)}{x} dx + \int_{\delta}^{\infty} \frac{f(x)}{x} dx \right\}$$

$$\int_{-\infty}^{\infty} \frac{x f(x)}{x^2 + \varepsilon^2} dx = P \left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx \right)$$

* Now we look at

$$i\varepsilon \int_{-\infty}^{\infty} \frac{f(x)}{x^2 + \varepsilon^2} dx = i\varepsilon \left(\int_{-\infty}^{-\delta} + \int_{-\delta}^{\delta} + \int_{\delta}^{\infty} \right) \left(\right)$$

again in

$$\rightarrow i \int_{-\infty}^{\infty} \frac{f(x)}{(x^2/\varepsilon) + \varepsilon} dx$$

$$x \notin [-\delta, \delta]$$

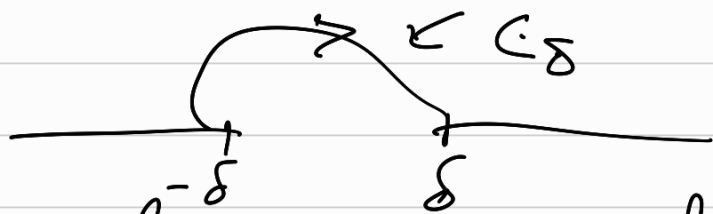
$$\lim_{\varepsilon \rightarrow 0} \left(\frac{f(x)}{(x^2/\varepsilon) + \varepsilon} \right) = 0$$

• So near $x \in [-\delta, \delta]$

$$\varepsilon f(0) \int_{-\infty}^{\infty} \frac{dx}{x^2} = \pi f(0)$$

Easier:

$$\int_c \frac{f(x)}{x} dx = P\left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx\right) + \int_{C_\delta} \frac{f(x)}{x} dx$$



$$\lim_{\delta \rightarrow 0} f(x) \rightarrow f(0)$$

$$\int \frac{f(x)}{x} dx = P\left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx\right) + \lim_{\delta \rightarrow 0} \int_{C_\delta} \frac{f(z)}{z} dz$$

$$f(z) \approx f(0)$$

$$z = \delta e^{i\theta} \Rightarrow dz = \delta i e^{i\theta} d\theta$$

$$(dz/z) = i d\theta$$

$$\lim_{\delta \rightarrow 0} \int_{\pi}^0 d\theta (f(0)i) = -\pi i f(0)$$

$$\int_c \frac{f(x)}{x} dx = P\left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx\right) - i\pi f(0)$$

$$\lim_{\epsilon \rightarrow 0} \int \frac{f(x)}{x \pm i\epsilon} dx = P\left(\int_{-\infty}^{\infty} \frac{f(x)}{x} dx\right) \mp i\pi f(0)$$

$$\left(\int f(x) \left[\lim_{\epsilon \rightarrow 0} \frac{1}{x \pm i\epsilon} - P\left(\frac{1}{x}\right) \pm i\pi \delta(x) \right] dx = 0 \right)$$

$$\therefore \lim_{\epsilon \rightarrow 0} \frac{1}{x \pm i\epsilon} = P\left(\frac{1}{x}\right) \mp i\pi \delta(x)$$

Thus,

$$\mathcal{F}_M[\hat{\Delta}(k^2)] = \pi \delta(k^2 + m^2) + \int_{4m^2}^{\infty} ds \rho(s) \pi \delta(k^2 + s)$$

$$= \pi \delta(k^2 + m^2) + \pi \rho(-k^2)$$

$$\rho(s) = 0, s < 4m^2$$

$$\mathcal{F}_M[\hat{\Delta}(-s)] = \pi \delta(m^2 - s) + \pi \rho(s)$$

$$\mathcal{F}_M[\hat{\Delta}(-m^2)] = \pi \delta(0) + \pi \rho(m^2)$$

$$\text{Im}[\tilde{\Delta}(-m^2)] = \pi \delta(0) + \pi \rho(m^2)$$

$$\boxed{\text{Im}[\tilde{\Delta}(-m^2)] = \pi \delta(0)} \quad (k^2 = -m^2)$$

(ii) $(s \geq 4m^2) \quad k^2 = -s \Rightarrow -k^2 \geq 4m^2$
 i.e. the total mass $M^2 \geq 4m^2$

$$\boxed{\text{Im}[\tilde{\Delta}(-s)] = \pi \rho(s)}$$

- Suppose $\text{Im}(\pi(k^2)) = 0$ for some range of k^2 , then

$$\tilde{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon} \Big|_{\text{Im}(\pi(k^2)) = 0}$$

$$\Rightarrow \text{Im}(\tilde{\Delta}(k^2)) = \pi \delta(k^2 + m^2 - \pi(k^2)) \quad , \quad \text{Im}(\pi(k^2)) = 0$$

From

$$\pi(-m^2) = 0$$

$$\Rightarrow \text{Im}(\tilde{\Delta}(-m^2)) = \pi \delta(0)$$

- $\text{Im}(\tilde{\Delta}(k^2)) = \pi \delta(k^2 + m^2) \quad , \quad \text{Im}(\pi(k^2)) = 0$
 $\rho(-k^2) = 0$ if $\text{Im}(\pi(k^2)) = 0$

Suppose $\text{Im}(\pi(k^2)) \neq 0$

$$\frac{1}{\alpha - i\epsilon} = \frac{\alpha + i\epsilon}{\alpha^2 + \epsilon^2}$$

$$\therefore \text{Im}(\tilde{\Delta}) = \frac{\epsilon}{\alpha^2 + \epsilon^2}$$

$$\alpha^2 = (k^2 + m^2 - \text{Re}(\pi))^2$$

$$\text{Im}(\tilde{\Delta}) = \frac{\text{Im}(\pi(k^2))}{(k^2 + m^2 - \text{Re}(\pi))^2 + \text{Im}(\pi)^2}$$

Comparing this with

$$\pi \rho(s) = \text{Im}(\tilde{\Delta}(-s)) \quad \boxed{s \geq 4m^2}$$

$$\boxed{\pi \rho(s) = \frac{\text{Im}(\pi(-s))}{(k^2 + m^2 - \text{Re}(\pi(-s)))^2 + \text{Im}(\pi(-s))^2}}$$

Since $\rho(s) = 0$ for $s < 4m^2$ then
 $\text{Im}(\pi(-s)) = 0$ for $s < 4m^2$

$$\text{Im} \{T(k^2)\} \approx 0 \quad \text{for} \quad k^2 > -4m^2$$